tinyAM :: CHEATSHEET

Formula-driven age-structured assessment

1 Setup tidy inputs

Use four tidy tables: `catch`, `index`, `weight`, and `maturity`. Each column is a variable; each row identifies a year and age, plus the survey for indices. Catch and biology need complete grids; survey rows may be sparse. Missing observations (`NA`) are allowed only in catch and index.

`catch`

year	age	obs	unit
1983	2	18	thousand fish
1983	3	2585	thousand fish
⋮	⋮	⋮	⋮
2025	19	0	thousand fish
2025	20	0	thousand fish

Optional `pop_covariates` table: year–age predictors on complete grids.

Catch/index: thousand fish. Weight: kg. Maturity: a proportion. `samp_time`: a fraction of the year.

`index`

year	age	survey	samp_time	obs	q_block	unit
1983	2	Fall 2J3KL	0.8	698822	2	thousand fish
1983	3	Fall 2J3KL	0.8	517528	3	thousand fish
⋮	⋮	⋮	⋮	⋮	⋮	⋮
2025	19	Fall 2J3KL	0.8	0.0	6-20	thousand fish
2025	20	Fall 2J3KL	0.8	0.0	6-20	thousand fish

`weight`

year	age	obs	unit
1983	2	0.0704	kg
1983	3	0.201	kg
⋮	⋮	⋮	⋮
2025	19	12.8	kg
2025	20	13.7	kg

`maturity`

year	age	obs
1983	2	0.00003
1983	3	0.00486
⋮	⋮	⋮
2025	19	1.00000
2025	20	1.00000

```
dat <- make_tam_data(
  catch = catch_data,
  index = index_data,
  weight = weight_data,
  maturity = maturity_data
)
```

2 Specify model hypotheses

These examples resemble statistical catch-at-age (SCAA), state-space assessment model (SAM), and bespoke Northern cod assessment model (NCAM) structures. They illustrate alternative assumptions using the same observations.

SCAA: fixed recruitment and mortality effects

Deterministic cohort survival; annual recruitment parameters; separable fishing mortality; fixed natural mortality.

```
scaa <- fit_tam(
  dat, years = 1983:2025, ages = 2:14,
  N_settings = list(process = "off", init = "exp"),
  rec_settings = list(process = "off", mu_form = ~ factor(year)),
  F_settings = list(process = "off",
    mu_form = ~ factor(year) + factor(age), mean_ages = 5:14),
  M_settings = list(process = "off",
    mu_supplied = ~ I(0.3), mean_ages = 5:14),
  catch_settings = list(sd_form = ~ 1),
  index_settings = list(sd_form = ~ 1, q_form = ~ mono(q_block))
)
```

SAM: stochastic abundance and fishing mortality

Abundance deviations, random-walk recruitment, and correlated random walks in fishing mortality; natural mortality stays fixed.

```
sam <- update(scaa,
  N_settings = list(process = "iid"),
  rec_settings = list(process = "rw", mu_form = NULL),
  F_settings = list(process = "cor_rw", mu_form = NULL)
)
```

NCAM: variation in fishing and natural mortality

Deterministic cohort survival, random-walk recruitment, and AR1 variation in fishing mortality and age-blocked natural mortality.

```
ncam <- update(scaa,
  rec_settings = list(process = "rw", mu_form = NULL),
  F_settings = list(process = "ar1", mu_form = ~ 1),
  M_settings = list(process = "ar1", age_breaks = c(2, 5, 8, 14))
)
```

Use `check_tam()` to inspect numerical diagnostics before interpreting estimates.

3 Visualize and compare

Explore model fits

```
vis_tam(scaa, sam, ncam)
```

Online dashboard: Interactive model fits, population patterns and residuals.

Test survey forecast skill

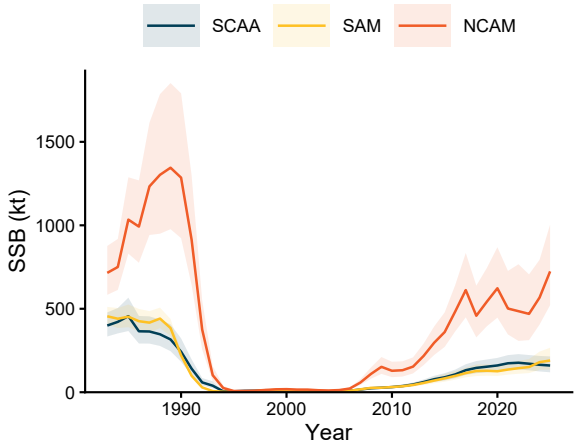
```
hind <- fit_hindcast(scaa, sam, ncam)
```

Assess survey forecast skill conditional on observed catches, and retrospective stability. Inspect `hind$fold_diagnostics` before comparing scores.

Access tables for report figures

```
out <- tidy_tam(scaa, sam, ncam)
ssb <- out$pop$ssb
index <- out$obs_pred$index
```

Population estimates and observation diagnostics are tidy tables, ready for static plots and report summaries.



Spawning stock biomass (kt). Lines show model estimates; shaded bands show 95% confidence intervals.

tinyAM :: MODEL REFERENCE

Settings, equations and the population assumptions they represent

Abundance and survival

N is start-of-year abundance. Each year, survivors move forward one age; survivors in the terminal age remain in the plus group. Let $Z = F + M$; recruitment is $R_t = N_{t,1}$.

$$\begin{aligned}\widehat{N}_{t,a} &= N_{t-1,a-1}e^{-Z_{t-1,a-1}},\\ \widehat{N}_{t,A} &= N_{t-1,A-1}e^{-Z_{t-1,A-1}}\\ &\quad + N_{t-1,A}e^{-Z_{t-1,A}},\\ \log N_{t,a} &= \log \widehat{N}_{t,a} + \eta^N_{t,a}.\end{aligned}$$

$\eta^N_{t,a}$ is the abundance deviation; see Variation in N, F and M for process choices. `init = "exp"` initializes older ages from exponential survivorship; other choices are `"free"`, `"latent"`, `"mono"`, and `"capped"`. Age indices $1, \dots, A$ run from the youngest modeled age to the plus group.
For $t \geq 2$, non-plus ages use the first survival equation.

Recruitment

`rec_settings: process = "off"` estimates a mean without process deviations; `mu_form = ~ factor(year)` gives annual recruitment. `"rw"` gives a random walk. With IID/AR1 residuals, `mu_form = ~ bh(ssb)` or `~ ricker(ssb)` adds a stock–recruit curve.

$$\begin{aligned}\text{RW: } \log R_t - \log R_{t-1} &= \epsilon_t,\\ \log R_t &= \log g(SSB_{t-L}) + u_t,\\ g_{BH}(S) &= \alpha S/(1+\beta S),\\ g_{Ricker}(S) &= \alpha S e^{-\beta S}.\end{aligned}$$

$\alpha, \beta > 0$; $\epsilon_t \sim \mathcal{N}(0, \sigma_R^2)$. IID has $u_t = \epsilon_t$; AR1 has $u_t = \phi_R u_{t-1} + \epsilon_t$ and a stationary start. Curves describe medians. Lag L defaults to the youngest biological age; early boundary R is free. No RW + curve.

Biomass and spawning biomass

Spawning biomass sums mature biomass across biological ages after mortality up to spawning.

$$SSB_t = \sum_b n_{t,b} w_{t,b} p_{t,b} e^{-\tau_{t,b} Z_{t,b}}.$$

Biomass is $\sum_b n_{t,b} w_{t,b}$. b: retained biological age; n: its abundance; w: weight; p: mature fraction; τ: year fraction from year start to spawning (default 0; set `maturity$spawn_time` for age/year-specific values). Hidden ages retain biology/timing and share plus-age mortality.

Mean fishing mortality

`F_settings: process = "off"` with `mu_form = ~ factor(year) + factor(age)` gives fixed separable year and age effects. `mu_form = ~ logistic(age)` gives a rising long-term mean.

$$\begin{aligned}S(a) &= \{1 + e^{-k(a-a_{50})}\}^{-1}, \quad k > 0,\\ \log F_{t,a} &= \beta_F + \log S(a) + \eta^F_{t,a}.\end{aligned}$$

With IID or AR1 process deviations, `mu_form = ~ mono(age)` instead adds cumulative nonnegative log-mean steps. Annual F can depart from the mean through η^F (see Variation in N, F and M).

Natural mortality

`M_settings: mu_supplied = ~ M0` supplies the baseline mortality surface.

$$\log M_{t,a} = \log M^0_{t,a} + \eta^M_{t,a}.$$

`M0` comes from `pop_covariates`. M states share age blocks and start in year 2 by default; other cells retain the mean. η^M denotes process deviations (see Variation in N, F and M); `mu_form` adds log-mean effects.

Variation in N, F and M

N and M support `"off"`, `"iid"`, `"rw"`, and `"ar1"`; F supports `"off"`, `"iid"`, `"rw"`, `"ar1"`, and `"cor_rw"`. The equations above use η^N, η^F, and η^M for these deviations.

$$\begin{aligned}\text{IID: } \eta_{t,a} &\sim \mathcal{N}(0, \sigma_a^2),\\ \text{RW: } \eta_{t,a} - \eta_{t-1,a} &\sim \mathcal{N}(0, \sigma_a^2).\end{aligned}$$

AR1 stationary covariance (state/block distances):

$$\text{Cov}(\eta_{t,a}, \eta_{s,b}) = \frac{\sigma_a \sigma_b \phi_t^{|t-s|} \phi_a^{|a-b|}}{(1-\phi_t^2)(1-\phi_a^2)}.$$

F `"cor_rw"`: independent annual increment vectors with

$$\Sigma_{ab} = \sigma_a \sigma_b \rho^{|a-b|}, \quad -1 < \rho < 1.$$

RW starts are unpenalized. `"off"` sets process deviations to zero.
N, F and M have separate process parameters. `sd_form = ~ 1` shares SDs; `~ factor(age)` gives age-specific SDs. `iid()`, `rw()` and `ar1()` terms structure log SD. Ordinary age formulas give $\sigma_a = \exp(X_a \beta_\sigma)$. Shared SDs do not share states.

Survey catchability

`q_block`: unrestricted age blocks; `mono(q_block)` imposes non-decreasing catchability under the default log link.

$$\eta_{q,k} = \beta_q + \sum_{j<k} d_j, \quad d_j \geq 0, \qquad q_k = e^{\eta_{q,k}}.$$

Zero steps give exact plateaus. With `q_link = "logit"`:

$$q_k = \{1 + e^{-\eta_{q,k}}\}^{-1}.$$

`~ logistic(age)` gives a logistic age pattern. Terms `iid()`, `rw()` and `ar1()` add Gaussian variation; q covariates come from `index`.

Catch and survey observations

Catch and survey observations are linked to population quantities by the predictions below; observed log values have Gaussian errors.

$$\begin{aligned}\widetilde{C}_{t,a} &= N_{t,a}\frac{F_{t,a}}{Z_{t,a}}(1 - e^{-Z_{t,a}}),\\ \tilde{I}_i &= q_i N_{t_i,a_i} e^{-s_i Z_{t_i,a_i}},\\ \log Y_i &\sim \mathcal{N}(\log \widetilde{Y}_i, \sigma_i^2),\\ \sigma_i &= \sigma_i^0 \exp(X_{sd,i}\beta_{sd}).\end{aligned}$$

s_i: survey sampling fraction. Catch/index `sd_form = ~ 1` shares an observation SD; `~ survey` allows survey differences. Baseline SDs and Gaussian formula effects can modify log SD. Predictions are medians; zero/NA observations are excluded, with complete predictions.

Fit and prediction checks

`check_tam()` checks numerical and statistical diagnostics; `fit_hindcast()` evaluates survey forecasts given observed removals and retrospective stability. The joint likelihood includes active process, formula-effect and observation densities. RTMB integrates random effects by Laplace approximation; intervals use local curvature. Standardized residuals $(\log Y_i - \log \widetilde{Y}_i)/\sigma_i$ depend on fitted, shrunken states. Convergence does not establish identifiability. Hindcasts return log RMSE and retrospective Mohn's ρ.